# Foundation Model Insights and a Multi-Model Approach for Superior Fine-Grained One-shot Subset Selection

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## Who should care?

- ML engineers training image classifiers under tight budgets.
- Practitioners facing fine-grained classes (pets, food, birds) and frequent dataset refreshes.
- Researchers seeking dataset-agnostic, one-shot subset selection with strong accuracy.



## Abstract

- **Goal**: reduce training cost via one-shot subset selection using information extractors (IEs).
- **Finding**: Foundation models (FMs) outperform traditional IEs on fine-grained datasets; gains diminish on coarse-grained noisy data.
- **Proposal**: RAM-APL combines multiple FMs to exploit complementary strengths.
- **Results**: SOTA on Oxford-IIIT Pet, Food-101, CUB-200-2011.

Figure placeholder: Pipelines comparison and performance overview



## Motivation: Dataset-Dependent IEs are Costly

- **Traditional pipeline**: pre-train IE on full target data before selection → extra compute and time.
- **Dataset dependency**: each new task requires re-pretraining (limits scalability).
- **Example**: TDDS needs 90 epochs on ImageNet-1K to gather dynamics, exceeding target training cost.

Visual: clock + GPU icons showing repeated pretraining per dataset

## Motivation: Compute and Time Costs (Quantified)

- **TDDS extractor**: 90 epochs on ImageNet-1K to log dynamics (paper: 90e).
- **Target model**: 60 epochs on coreset can be less than extractor cost (paper remark).
- **FM-based**: 0 epochs task-specific pretrain; reuse features for selection.

Table/callouts: GPU-hours and wall-clock under fixed hardware (illustrative placeholders aligning with 90e vs 60e; FM pretrain: 0e task-specific)



## Motivation: Real-world Pain Points

- Large-scale datasets make full pretraining prohibitive (compute, time).
- Noisy/coarse-grained labels reduce benefit from traditional IEs.
- Frequent task/domain shifts require repeated extractor training.

Examples: new product categories; frequent dataset refreshes; limited GPU budgets



## Motivation: Consolidated Timeline & Cost Burden

Unified timeline: $D_1 \ldots D_4$ over months — Traditional: IE retrain 90e each cycle; FM: single FM → measure/select; Multi-FM (CLIP+DINOv2) → RAM-APL → train

- Traditional: repeated extractor pretraining dominates cycle time.

## Motivation Case: E-commerce Fine-Grained Catalog

- Weekly new SKUs with fine-grained classes → constant domain shifts.
- Traditional IE: retrain per refresh; delays launches and inflates compute budgets.
- RAM-APL: reuse CLIP+DINOv2 features; select 30–50% data for training.
- **Impact**: cut extractor pretraining (90e) to 0; train on subset with gains reported in paper.

Timeline: Week t — Traditional (IE retrain 90e + select + train) vs RAM-APL (FM reuse 0e + select + train on 30–50%)

## Why Foundation Models? Dataset-Agnostic Reuse

- **Reusable features**: DINOv2, CLIP used across tasks without task-specific pretraining.
- **Decoupled selection**: subset choice without full-dataset fine-tuning.
- **Generality**: strong on fine-grained datasets per paper experiments.

Visual: FM feature extractor reused across multiple datasets



## Why FMs? Concrete Efficiency Wins

- **Pretraining saved**: 0 task-specific epochs vs traditional IE training (e.g., 90e for TDDS).
- **Subset training**: select 10–70% with class balance; retain accuracy gains over Random and baselines.
- **Multi-FM**: RAM-APL with {CLIP, DINOv2} yields best trade-off.

Callouts with numbers: 90e → 0e; sampling $p \in \{10,30,50,70\}\%$



## Motivation: Selecting the Right FM is Hard

- **Variation**: Different FMs excel under different datasets/methods/rates.
- **Mismatch**: High zero-shot accuracy ≠ best subset selection.
- **Need**: multi-FM approach that fuses complementary strengths.

Visual: heatmap of best-FM-by-setting



## Problem Framing

- **Subset (coreset) selection**: pick a small informative subset S from dataset D under budget p.
- **Focus**: one-shot methods (single pass) vs. adaptive (iterative).
- **Challenge**: traditional, dataset-dependent IEs require costly target pretraining.

Figure 1(a): Traditional pipeline schematic

## Preliminary & Setup (Compact)

- Goal: select subset S at budget p to match full-data performance; we evaluate across standard fine/coarse datasets with multiple IEs and selection methods per paper.



## Single-Model Study: Key Findings

- **Obs1**: Limited FM advantage on noisy, coarse-grained datasets (e.g., CIFAR-10N).
- **Obs2**: Strong, consistent gains on fine-grained datasets (Pet, Pet-N).
- **Obs3**: Different FMs rank differently across datasets/methods/rates; higher zero-shot accuracy ≠ better selection.

Composite visual: frequency of best IE across setups + scatter of FM downstream acc vs selection



## Proposed Method: RAM-APL (Schematic)

- **Multi-FM IEs**: use CLIP-VITL14 and DINOv2-VITs14 features.
- **RAM (intra-class)**: per-FM class centers → distance ranks → mean rank across FMs.
- **APL (inter-class)**: nearest-center pseudo-label per FM → average correctness $\varphi_j$.
- **Score**: $W_1(p) \cdot RAM + W_2(p) \cdot (1-\varphi)$; select smallest up to budget; $\alpha=0.2$, $\beta=1$.

Single infographic: CLIP+DINOv2 → RAM ranks + APL pseudo-labels → weighted score → selection



## Results & Conclusion

- **Outperforms baselines** across sampling rates on Pet, Food-101, CUB-200-2011.
- **Avg gain over Random**: +3.74% (Pet), +4.44% (Food-101), +6.40% (CUB-200-2011).
- **Takeaway**: FMs excel as IEs on fine-grained data; RAM-APL leverages multi-FM fusion for SOTA one-shot selection.

Accuracy vs sampling rate curves for Pet, Food-101, CUB-200-2011 with annotations for RAM-APL improvements and bar insets of average gains

## References & Resources


- Radford et al., 2021 (CLIP)
- Oquab et al., 2023 (DINOv2)
- [Welling09] Welling, 2009 (Herding)
- [Sener17] Sener & Savarese, 2017 (K-Center Greedy)
- [Iyer21] Iyer et al., 2021 (Graph Cut)
- [Agarwal20] Agarwal et al., 2020 (Contextual Diversity)
- [Coleman19] Coleman et al., 2019 (Margin / Proxy)
- [Toneva19] Toneva et al., 2019 (Forgetting)
- [Paul21] Paul et al., 2021 (GraNd)
- [Margatina21] Margatina et al., 2021 (Cal)
- [Killamsetty21b] Killamsetty et al., 2021b (Glister)
- [Xia23] Xia et al., 2023 (MDS)
- Wei et al., 2022 (CIFAR-10N); Cui et al., 2019 (Imbalance)
- Parkhi et al., 2012 (Pet); Bossard et al., 2014 (Food-101); Wah et al., 2011 (CUB-200-2011)

- **Code**: github.com/ZhijingWan/RAM-APL
- More details in Appendices A–B (paper).



## Appendix: Datasets, IEs, and Training

- Datasets: CIFAR-10, CIFAR-10N, CIFAR-10I, Oxford-IIIT Pet, Pet-N; fine-grained eval: Pet, Food-101, CUB-200-2011.
- IEs: model-TD (10e target), model-TIN (10e TinyImageNet), single FM (DINOv2/CLIP/SigLIP/EVA-CLIP); RAM-APL uses CLIP-L/14 + DINOv2-S/14.
- Selection: MIN, KCG, GC, MDS; $p \in \{10\%, 30\%, 50\%\}$. Targets: ResNet-18/50; SGD with cosine decay; standard augmentations as in paper.